Guidelines for Final Review

Individual Technical Competency and Conceptual Understanding Assessment

1. Objective of the Review

The purpose of this review is to assess each student's individual understanding of the project undertaken by the group. Every student is expected to demonstrate a clear understanding of the theoretical concepts, algorithms, system architecture, design decisions, implementation, and results relevant to their project.

The review will focus on **understanding, reasoning, and the ability to explain technical concepts independently**, rather than on the quality of the presentation slides alone.

2. Instructions to Students

A. Understand the Project Thoroughly

Every student must be familiar with the complete project, not just the module or task assigned to them. You should be able to explain:

- The problem being addressed and its significance.
- The objectives and expected outcomes of the project.
- The theoretical foundations and technical concepts used.
- The system architecture, design methodology, and workflow.
- The algorithms, models, data structures, or techniques employed and the reasons for selecting them.
- The implementation approach, testing methodology, and results obtained.
- The limitations of the proposed solution and possible improvements.

B. Be Prepared for Individual Questioning

- Each student will be questioned individually, irrespective of their role in the project.
- Questions may be asked from any part of the project, including the theoretical concepts on which it is based.
- You may be asked to justify why a particular algorithm, model, framework, or technology was selected over alternative approaches.
- You should be able to explain the underlying concepts in your own words without depending on the presentation slides.
- You may be asked to interpret results, identify limitations, analyse a failure case, or suggest modifications to the solution.

C. Whiteboard Explanation is Mandatory

Students should be prepared to explain concepts and solve small technical problems on the whiteboard when asked by the faculty.

Depending on the nature of the project, you may be asked to:

- Draw and explain the system architecture or workflow.
- Illustrate an algorithm step by step using a suitable example.
- Write pseudocode or explain the logic of a critical section of the program.
- Explain a mathematical equation, model, or theoretical principle used in the project.
- Trace the execution of an algorithm or explain the processing of a sample input.
- Draw a relevant data structure, diagram, flowchart, or sequence of operations.
- Explain how data moves through the system and how the output is generated.
- Analyse a simple modification to the problem and explain how your solution would respond.

The explanation must demonstrate your understanding of the underlying concepts, not merely reproduce a diagram or definition from memory.

D. Demonstrate Individual Contribution

Each student must clearly explain their own contribution to the project and demonstrate that they understand the work they claim to have completed.

However, individual contribution does not exempt a student from questions on the overall project, its fundamental theory, or its major design decisions.

3. Use of AI Tools

AI tools may be used as supporting resources for literature exploration, coding assistance, debugging, documentation, and presentation preparation, subject to institutional guidelines.

However:

- AI-generated content must be reviewed and understood by the students before it is included in the project or presentation.
- Students must be able to explain and defend the technical content they present.
- Copying definitions, diagrams, algorithms, code, or explanations without understanding them will not demonstrate technical competency.
- Students must be prepared to verify, correct, or critically evaluate suggestions made by AI tools.

The assessment will focus on your actual understanding of the work, irrespective of the tools used to prepare it.

Every student must be prepared to explain the project independently, without relying on slides, notes, or another team member's explanation.